

Disk Drive Performance Activity

Short Answer

1. In your own words, define seek time.
2. Why does random I/O suffer more from seek time than sequential I/O?
3. What is short-stroking, and why would an administrator use it?

Problem Solving (Capacity Loss from Short-Stroking)

1. A 1 TB (1000 GB) disk drive is short-stroked to use only the first 30% of its cylinders.
 - a) What is the usable capacity of the short-stroked drive?
 - b) How much capacity is "wasted" (unusable)?
2. A disk drive has:
 - Full-stroke seek time: 20 ms
 - Average seek time (full drive): 8 ms
 - Average seek time (short-stroked to 40%): 3 ms
 - Rotational latency: 2 ms
 - Data transfer time: 0.5 ms

Calculate the service time and IOPS for each configuration. (Show solution and formula used)

Configuration	Service Time (ms)	IOPS
Full Drive		
Short-Stroked (40%)		

Compare

1. An administrator has two identical 600 GB drives:

Drive A: Full drive (all cylinders used)

Drive B: Short-stroked to use only 50% of cylinders

Question	Drive A	Drive B
Usable capacity		
Relative seek distance (Long / Short)		
Expected IOPS (Higher / Lower)		
Best suited for		